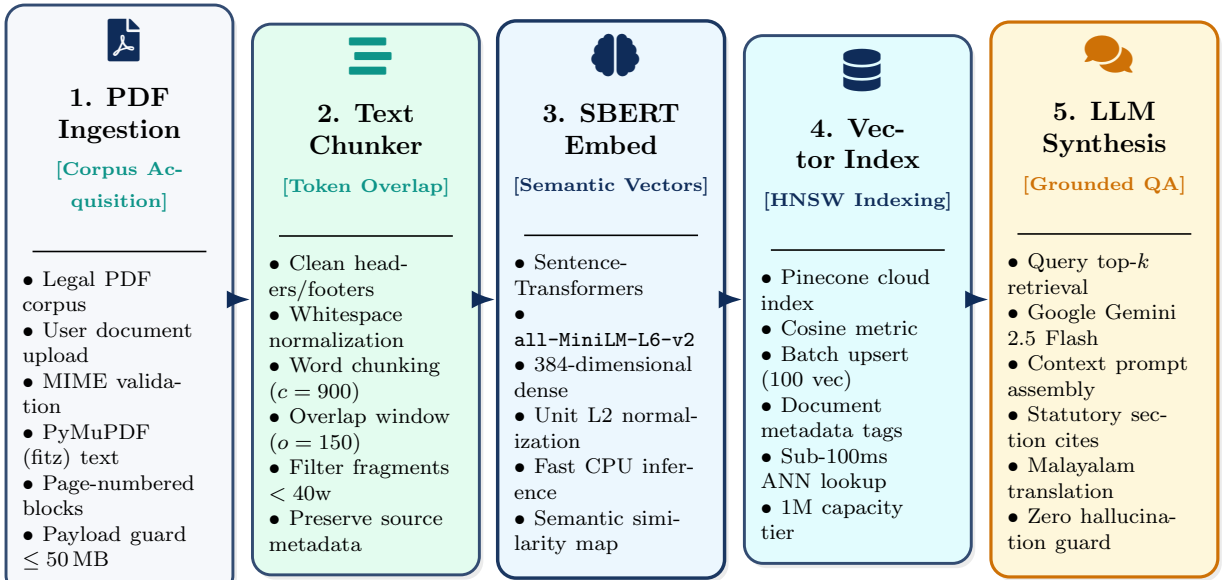

1. PDF Ingestion
[Corpus Acquisition]
• Legal PDF corpus
• User document upload
• MIME validation
• PyMuPDF (fitz) text
• Page-numbered blocks
• Payload guard ≤ 50 MB
2. Text Chunker
[Token Overlap]
• Clean headers/footers
• Whitespace normalization
• Word chunking ($c = 900$)
• Overlap window ($o = 150$)
• Filter fragments < 40w
• Preserve source metadata
3. SBERT Embed
[Semantic Vectors]
• Sentence-Transformers
• all-MiniLM-L6-v2
• 384-dimensional dense
• Unit L2 normalization
• Fast CPU inference
• Semantic similarity map
4. Vector Index
[HNSW Indexing]
• Pinecone cloud index
• Cosine metric
• Batch upsert (100 vec)
• Document metadata tags
• Sub-100ms ANN lookup
• 1M capacity tier
5. LLM Synthesis
[Grounded QA]
• Query top-k retrieval
• Google Gemini 2.5 Flash
• Context prompt assembly
• Statutory section cites
• Malayalam translation
• Zero hallucination guard